G1 · Experiment 2 test plan · v0.3 · 10 November 2026

Status: draft for Renxi. Follows «Experiment 2 Context» (Renxi, 10 Nov), which governs where this plan and that document differ. Nothing in stages 2 to 5 is built or run yet. No threshold, tolerance or rule is fixed in this document.

1. Objective

The G1 part of Experiment 2 is a **cross-embodiment reconstruction test of CCM**, not a test of doorway-crossing performance.

Concept (Renxi, 10 Nov): directional pre-entry compatibility is the functional relationship among doorway geometry, the G1 passage envelope and posture, stop distance, lateral offset, heading, proposed motion and sensing validity that determines whether the crossing capability is sufficiently supported for participation.

Primary question: can CCM reconstruct directional pre-entry compatibility under the G1 embodiment when pose, lighting and blind-spot conditions vary, with C2.5 returning the appropriate *within span, boundary, outside span or indeterminate* and C3 the appropriate *participate, review, defer, no-use or renew*?

Decisive comparison: the fixed 0.85 m and 8° heuristic versus CCM reconstruction of embodied and sensing-dependent compatibility. CCM adds value only where the fixed heuristic is insufficient.

Two parallel tracks (Renxi, 10 Nov). Summit XL tests the concept under a simulation-to-physical context change within the same embodiment. G1 borrows the **abstract concept** and tests whether it can organise and explain observations under a different embodiment. The tracks are complementary, not sequentially dependent: the G1 pre-test proceeds as soon as its own frozen-record mechanism and evidence schema are valid, and does not wait for the Summit matched records.

G1 imports nothing Summit-specific: no thresholds, classifications, clearances, sensing assumptions or authority rules. The concept's local meaning is reconstructed from G1 geometry, posture, passage envelope, proposed motion, lighting, blind spots, localisation and perception validity.

Motion-routine performance is not concept-reconstruction performance. The routine is only the operational means of reaching the aligned decision point; the experiment begins at that point. No crossing time, clearance, collision count or completion rate is collected as experimental data.

Crossing outcome is supporting operational evidence, not the primary ground truth. An appropriate defer is not a failure because the robot might have crossed.

2. Stages

Stage («Experiment 2 Context»)	What it produces on G1	Data status	State today
1. Development characterisation	Acquisition feasibility, achieved pose variation, routine behaviour, sensing limitations, missing record fields, safety constraints	Development	Ready to run (session manual v4)
2. Frozen-record validation	A validated frozen record at `DOOR-ENG aligned`	Pre-test	Not built
3. Controlled pre-test	Frozen records for a selected set of concept-boundary cases	Pre-test	Not laid out

Stage («Experiment 2 Context»)	What it produces on G1	Data status	State today
4. Lock	Frozen concept definition, applicability and uncertainty conditions, span categories, authority boundaries, prompts, model versions, preprocessing, thresholds, analysis rules	Lock	Not reached
5. Locked G1 evaluation	New G1 records; C2, C2.5 and C3 compared with nothing altered after inspecting outputs	Confirmatory	Starts only after 2, 3 and 4; no confirmatory claim before

Development and pre-test records are never reused as confirmatory evidence. G1 records are not pooled with Summit XL records.

3. Stage 1 · Development characterisation

The 12 declared crossings (`DEV_r01` to `DEV_r12`, seed 202610011), existing routine, existing safety conditions.

Characterised: acquisition feasibility, achieved stop distance, lateral offset and heading, direction of travel, posture and carried-object state, routine behaviour, contact observations, sensing limitations, missing record fields, operational safety constraints.

Already available from the existing G1 history (real runs, pose at the aligned point): A to B, n = 23, stop distance 1.04 m (sd 0.08), heading within about ±8°; B to A, n = 27, 0.98 m (sd 0.05), heading from −23° to +8°. These describe the natural variation of the aligned state under the existing heuristic. They are development figures, not evidence about the concept.

Not recorded today by the stack: posture and carried-object state, and lighting. They go on the observer sheet until stage 2 records them.

4. Stage 2 · Frozen-record acquisition pre-test

One immutable record written when the stack logs `DOOR-ENG aligned`, robot standing still, before the crossing starts.

Field (Renxi's list)	Source on G1	State
Scan or relevant perception data	Robot laser at the aligned point	To be taken from the stack; topic to confirm
Pose and covariance	Localisation of the common stack	To confirm that covariance is published
Transforms	TF tree at the cut	To confirm completeness
DKP projection	Same projection code as Summit XL, with G1 parameters	Not built
Estimated doorway geometry	From the scan at the cut	Not built for G1
G1 posture and passage envelope	Joint state and declared envelope	**Needs Renxi: how the envelope is declared**
Stop distance, lateral offset, heading	Computed from pose and door frame	Computation exists (history script)
Proposed crossing direction and motion	From the goal and the crossing skill	Available in the stack
Lighting metadata	Category plus instrumented measure where available	Not built

Field (Renxi's list)	Source on G1	State
Blind-spot metadata	Which required evidence is unavailable	Not built
Perception-validity indicators	Data fresh, transforms complete, confidence sufficient, required evidence present	Not built
Manifest: ContextID, PairID, CutID, TraversalID, t_aligned, t_freeze, component timestamps, file hashes	Valid-cut manifest, same format as Summit XL	Manifest exists for Summit XL (10 Nov, bench-tested)

Validation before any further collection: timestamp consistency, pose freshness, transform completeness, record integrity. Reported as attempted, aligned, frozen, valid.

Order of work: build and validate in the G1 twin first; then one real acquisition smoke test with Natan present. The frozen record sends no command to the robot.

5. Stage 3 · Controlled concept-boundary pre-test

Frozen records under controlled variation in: stop distance, lateral offset, heading, direction (A to B, B to A), posture or carried object, lighting (reproducible categories: nominal, reduced, asymmetric, glare or strong contrast, with an instrumented measure where available), blind spot (stated as which required evidence is unavailable: left or right doorway boundary, robot-to-frame clearance, localisation reference, transform path, passage-envelope evidence).

The set deliberately includes each conceptual outcome:

Scenario	Geometry	Evidence	Expected C2.5	Expected C3
Clearly supported	Compatible	Valid	Within span	Participate
Marginal relation	Near boundary	Valid	Boundary	Review
Conflicting evidence	Apparently compatible	Conflicting	Boundary or indeterminate	Review
Missing evidence	Apparently compatible	Insufficient	Indeterminate	Defer
Unsupported relation	Incompatible	Valid	Outside span	No-use
Structural mismatch	Adequately observed	Source concept inadequate	Outside span	Renew
Supported but unauthorised	Compatible	Valid	Within span	Participate; action status: not authorised

Not a full factorial: a balanced set of deliberately selected, safe and reproducible cases.

Not every record ends in a crossing. Defer and no-use conditions stop at the aligned point. The record is frozen and the robot is walked back.

Open for Renxi before this stage is laid out: which cases, how many records per case, and who writes the reference annotation for each record and when.

6. Offline reconstruction (applies in stages 3 and 5)

C2, C2.5 and C3 evaluate the same immutable records, offline, with no command to the robot.

- **C2** reconstructs the observable relationships among geometry, embodiment, pose, sensing and proposed motion.
- **C2.5** returns whether the record lies within the contextual span: **within span, boundary, outside span, indeterminate**. It is read from the C3 record before the additional CCM layers: a paired intermediate output (C2.5 is contained in C3), not an independent model run.

- **C3** adds G1-local meaning, unresolved conditions, possible-role relationships and authority, and returns **participate, review, defer, no-use or renew**. Missing evidence gives indeterminate at C2.5 and defer at C3, not a forced classification.
- **Action authority is a separate field** from the C3 outcome: a record may return *participate* while its action status is *not authorised*. Resolution or eligibility never grants permission to act.
- **Renew** means a *candidate* specification renewal: it must still be grounded, validated and released, and never silently replaces the existing concept.
- **Both output layers** are adapted to these exact sets, C2.5 and C3.
- **Two C3 calls**, rule fixed in advance: the first is the primary result, the second is a repeatability assessment, disagreement is reported as instability.

Reported separately: attempted, aligned, frozen, valid, classified.

Primary outcomes (none is crossing success): relation reconstruction, contextual-span correctness, governance correctness, unresolved-condition fidelity, embodiment sensitivity, proposal control, authority separation, repeatability, incremental contribution over C2 and over the fixed heuristic.

7. Stages 4 and 5 · Lock and locked G1 evaluation

Locked after the pre-test and before any confirmatory record: concept definition, record schema, thresholds, prompts and model versions, classification rules, reference annotations, outcome metrics.

Then **new records** are collected. Confirmatory evaluation begins only after the aligned frozen-record acquisition of stage 2 has been implemented and validated.

8. Safety, unchanged

Nothing runs on G1 without Natan beside the robot. The stack does not rescue itself. Stop conditions are those of the session manual. Degraded-lighting and blind-spot conditions are set up with the robot standing still.

9. Questions for Renxi

1. Passage envelope and posture: declared once per configuration, or derived per record from the joint state?
2. Reference annotation: who writes the expected C2.5 and C3 result of each record, and before or after seeing the record?
3. Confidence in the perception-validity indicator (“confidence sufficient”): which quantity on G1 is to be used?